SURROGATE-ASSISTED DOUBLE MACHINE LEARNING: A UNIFORM INTERFACE FOR DISCRETE OUTCOMES

A PREPRINT

Brandon M. Greenwell
Department of Operations, Business Analytics, and Information Systems
University of Cincinnati Carl H. Lindner College of Business
Cincinnati, Ohio, 45221
greenwbm@ucmail.uc.edu

Other Authors
University of XYZ
other.authors@email.com

2026-08-16

ABSTRACT

Regression on a discrete label generally targets the wrong scale for a cumulative-link or log-mean effect. Surrogate-Assisted Double Machine Learning (SUDO) first completes each observed category to a continuous surrogate using a fitted conditional CDF, then applies ordinary partialling-out. If the fitted distribution is correct, randomized probability-integral-transform completion restores the scalar treatment index in conditional mean. Repeated completions with proper multiple-imputation draws provide inference for parametric models; a full-pipeline bootstrap serves the same role for flexible learners. Simulations show nominal coverage in the validated binary and ordinal designs and also expose the method's main limitation: error in the imputation distribution is not removed by DML orthogonality. For a coefficientless penalized additive learner, scalar recalibration followed by SUDO removes treatment-regularization bias in three additive designs; a high-dimensional boundary fails. SUDO matches, rather than dominates, tailored binary estimators. Its clearest use is for ordinal outcomes, where the same construction targets a latent proportional-odds coefficient without a new outcome-specific score. A wine quality application illustrates the method. An oracle experiment validates count completion, while fitted count learners remain unpromoted.

1 Introduction

Many outcomes are discrete: relapse is binary, survey responses are ordinal, and trips are counts. Double Machine Learning (DML, Chernozhukov et al. 2018) is well suited to a continuous partially linear outcome because it removes flexibly estimated covariate effects and reads the treatment effect from a residual-on-residual regression. Applying the same calculation to a discrete label is easy, but it usually targets a residualized contrast in $E[Y \mid D, X]$. It does not recover a cumulative-link coefficient or a log conditional-mean effect. The problem is the outcome scale.

This paper takes a simple route around that mismatch: complete the observed category to a continuous surrogate on the target scale, then run the usual partialling-out calculation. The completion uses the fitted conditional CDF to draw within the jump corresponding to the observed category. Under a correct distribution, that randomized probability-integral transform has a known continuous reference law. Adding the fitted scalar index produces a surrogate whose conditional mean is the desired partially linear index.

The construction repurposes the surrogate-variable approach of D. Liu and Zhang (2018), originally developed for residual diagnostics in ordinal regression, as an estimation device. Cheng, Wang, and Zhang (2021) extend that approach to unordered discrete-choice models with vector utilities, while Greenwell et al. (2018) provide its binary and ordinal implementation in the `sure` package. SUDO stays in the scalar-index setting: a distribution learner supplies a pointwise index and conditional CDF, and SUDO supplies the completion, partialling-out, and uncertainty calculation. Binary, ordinal, and count laws then share one estimator interface.

That uniformity does not make SUDO uniformly better. Tailored orthogonal scores exist for logistic binary models (M. Liu, Zhang, and Zhou 2021) and Poisson log-mean effects (Drukker and Liu 2022). In our binary comparisons, SUDO matches the correct tailored estimator and direct coefficient rather than improving on them. The main practical opening is ordinal regression, where we identify no published orthogonal-score analogue for the latent proportional-odds coefficient.

The paper makes three points. First, randomized-CDF completion turns a broad class of discrete outcomes into an ordinary partially linear problem. Second, inference must propagate completion and imputation-model uncertainty: proper multiple-imputation draws do so for parametric models, while a full-pipeline bootstrap does so for validated flexible learners. Third, DML orthogonality protects the adjustment regressions but not the distribution used to create the surrogate. Simulation therefore remains essential when a new imputation learner is introduced. We validate the binary and ordinal implementations, illustrate the ordinal method with wine quality, and report an oracle count bridge without promoting the fitted count analysis.

2 The surrogate idea

Let D be a treatment, $X \in \mathbb{R}^p$ covariates, and Y a discrete outcome with conditional CDF $G_0(y \mid D, X)$. The true distribution has a scalar additive index

$$\eta_0(D, X) = \theta_0 D + g_0(X). \tag{1}$$

For binary and ordinal outcomes this is a latent-utility index: binary $Y = \mathbf{1}\{Z > 0\}$, or ordinal, $Y = j \iff \kappa_{j-1} < Z \le \kappa_j$ for thresholds $-\infty = \kappa_0 < \kappa_1 < \cdots < \kappa_{J-1} < \kappa_J = \infty$. The latent utility follows a partially linear model

$$Z = \theta_0 D + g_0(X) + \varepsilon, \qquad \varepsilon \mid D, X \sim F \tag{2}$$

with g_0 an arbitrary nuisance function and F a known error distribution (logistic for proportional odds, extreme-value for complementary log-log, and so on). The target θ_0 is the treatment effect on this latent scale, with g_0 left nonparametric. It is not a probability-scale average treatment effect. Under a logistic link, $\exp(\theta_0)$ is a proportional-odds ratio. For Poisson or negative-binomial counts we use $E[Y \mid D, X] = \exp\{\eta_0(D, X)\}$, making θ_0 a log conditional-mean effect and $\exp(\theta_0)$ a conditional rate ratio.

This notation matches the surrogate literature up to sign convention. D. Liu and Zhang (2018) write the ordinal latent variable as $Z = -f(X, \beta) + \varepsilon$, draw the surrogate from $S \mid (Y, X) \sim Z \mid (Y, X)$, and define the surrogate residual $R^S = S - E_0(S \mid X)$. Here $\eta = -f$, treatment is included in the index, and the reference error is centered, so the corresponding correct-model residual is $R^S = S - \eta(D, X)$. SUDO uses the surrogate S as an outcome for estimation rather than using R^S only for diagnostics. The vector surrogate of Cheng, Wang, and Zhang (2021) is broader, but is not required for the scalar estimands studied here.

Identification requires four conditions.

1. *Selection on observables.* No unobserved confounder acts beyond X. Together with the conditional location-family statement in Equation 2, this assumption requires the latent error to have the same normalized law across treatment values given X.
2. *A constant index-scale effect.* The partially linear form in Equation 1 makes θ_0 a single number rather than a function of X.
3. *A correctly specified discrete conditional distribution.* Completion uses the entire fitted CDF jump, including the link and thresholds for cumulative models or the family and dispersion for counts.
4. *Overlap.* Treatment retains support after adjustment, so its residual is not annihilated.

The first and fourth are standard causal conditions. The second and third are modeling choices shared with any structured distributional analysis.

2.1 From a category to a continuous outcome

FWL estimates a treatment coefficient by regressing an outcome residual on a treatment residual after removing X from both. DML combines that identity with cross-fitting and flexible nuisance regressions (Chernozhukov et al. 2018). SUDO changes only the outcome supplied to this calculation.

Fit a structured distributional model of Y on (D, X). It must return the scalar index $\hat{\eta}_i$, the fitted CDF immediately below and at the observed outcome, and any global distribution parameters. Draw

$$V_i \mid Y_i, D_i, X_i \sim \mathrm{Unif}\{\hat{G}(Y_i - 1 \mid D_i, X_i), \hat{G}(Y_i \mid D_i, X_i)\}, \qquad S_i = \hat{\eta}_i + H^{-1}(V_i), \tag{3}$$

where H is a continuous mean-zero reference law. For binary and ordinal cumulative links, we set H to the centered link-error law. This gives the truncated latent-variable surrogate of D. Liu and Zhang (2018), expressed through its CDF jump, up to the corresponding intercept centering. For counts we use standard normal H, following the randomized-quantile-residual convention of Dunn and Smyth (1996). The reference law changes the completion variance but not the target.

For the normal count reference, the completion has an analytic Rao-Blackwell form. With $a = \hat{G}(Y-1 \mid D, X)$, $b = \hat{G}(Y \mid D, X)$, and standard-normal density ϕ,

$$E\{\Phi^{-1}(V) \mid Y, D, X\} = \frac{\phi\{\Phi^{-1}(a)\} - \phi\{\Phi^{-1}(b)\}}{b-a},$$

with the continuous limiting value for a zero-width numerical jump. The count stages report this analytic completion beside the $B = 25$ randomized completion.

Proposition 1 (randomized-CDF completion). *If $G = G_0$ and $\eta = \eta_0$, and if H is continuous with $E\{H^{-1}(V)\} = 0$ for $V \sim \mathrm{Unif}(0,1)$, then the completion in Equation 3 satisfies $E[S \mid D, X] = \eta_0(D, X)$. Consequently,*

$$S = \theta_0 D + g_0(X) + \eta, \qquad E[\eta \mid D, X] = 0$$

and population FWL recovers θ_0.

Proof. Conditional on (D, X), each discrete outcome selects one disjoint CDF jump with probability equal to that jump's width. Uniform randomization inside the selected jump therefore makes $V \mid D, X$ uniform on $(0,1)$. The reference quantile has conditional mean zero, leaving $\eta_0(D, X)$. $\square$

For cumulative links, this is the conditioning identity behind the surrogate-variable results of D. Liu and Zhang (2018) and Cheng, Wang, and Zhang (2021). The estimation step is to use S, rather than only its diagnostic residual, as the outcome in FWL or DML.

With an estimated distribution, the equality is asymptotic under consistency and approximate otherwise. This limitation matters because the imputation model is not protected by DML orthogonality.

The following base R example compares the fitted logistic coefficient, FWL on the unobserved utility, and FWL on one surrogate completion.

```r
set.seed(1)
n <- 40000
theta <- 1.5
X <- rnorm(n)
D <- rbinom(n, 1, plogis(0.8 * X))
Z <- theta * D + X + rlogis(n)
Y <- as.integer(Z > 0)

imputation_fit <- glm(Y ~ D + X, binomial)
index_hat <- predict(imputation_fit)
cdf_at_zero <- plogis(-index_hat)
uniform_draw <- ifelse(
  Y == 1,
  cdf_at_zero + runif(n) * (1 - cdf_at_zero),
  runif(n) * cdf_at_zero
)
surrogate <- index_hat + qlogis(
  pmin(pmax(uniform_draw, 1e-9), 1 - 1e-9)
)

treatment_residual <- resid(lm(D ~ X))
fwl <- function(outcome) {
  sum(treatment_residual * resid(lm(outcome ~ X))) /
    sum(treatment_residual^2)
}
round(c(
  glm = unname(coef(imputation_fit)["D"]),
  oracle = fwl(Z),
  surrogate = fwl(surrogate)
), 3)
```

```
  glm    oracle surrogate
1.497     1.496     1.505
```

All three estimates are close to $\theta_0 = 1.5$. Using only the binary labels, the surrogate completion recovers the coefficient that FWL would estimate if the latent utility were observed. Replacing the binary index with an ordinal cumulative-link index and truncating to category-specific intervals gives the ordinal construction. The remainder of the paper adds two elements to this basic estimator: machine-learning partialling and inference that accounts for the uncertainty hidden by a single completion.

3 Estimation and inference

The estimator. Cross-fit the two adjustment nuisances $\widehat{m}(X) \approx E[D \mid X]$ and $\widehat{\ell}(X) \approx E[S \mid X]$, and form the residual-on-residual estimate

$$\widehat{\theta} = \frac{\sum_i (S_i - \widehat{\ell}(X_i))(D_i - \widehat{m}(X_i))}{\sum_i (D_i - \widehat{m}(X_i))^2} \tag{4}$$

The $\widehat{\ell}$ regression uses X only: the index carried D's signal into S, and partialling on X alone isolates θ_0. The score $\varphi = (S - \ell(X) - \theta(D - m(X)))(D - m(X))$ is Neyman-orthogonal in (ℓ, m): its derivatives in both nuisance directions vanish by the tower property, so first-order error in $\widehat{\ell}$ or $\widehat{m}$ does not bias $\widehat{\theta}$. This protection covers the two adjustment nuisances, not the imputation model that produced the surrogate.

Proper multiple-imputation inference. A single completion understates uncertainty, so we draw B surrogates and pool with Rubin's rules (Rubin 1987): the estimate is $\bar{\theta} = B^{-1} \sum_b \widehat{\theta}^{(b)}$ and the total variance is $T = \bar{W} + (1 + 1/B)B_*$, with a Barnard-Rubin small-sample t reference (Barnard and Rubin 1999). Reusing one fitted $\widehat{\beta}$ across all completions leaves its shared estimation error invisible to the between-draw variance. A *proper* draw redraws the imputation model's parameters, $\beta^{(b)} \sim N(\widehat{\beta}, \widehat{\mathrm{Cov}}(\widehat{\beta}))$ (thresholds included for ordinal outcomes), before each completion. This is the large-sample normal form of a proper posterior-predictive draw. It accounts for sampling noise, surrogate randomness, and imputation-model uncertainty. Table 1 shows why all three matter.

For ordinal outcomes, thresholds are parameters too and must be perturbed jointly with the coefficients. Surrogate-residual mean and variance checks provide a diagnostic for the assumed link and mean structure (D. Liu and Zhang 2018), but they do not replace target-level validation.

3.1 What completion adds to the fitted index

Let $R_D = D - m_0(X)$ and let a fitted distribution learner return the pointwise index $\widehat{\eta}(D, X)$ without requiring a treatment coefficient. Write $\mu_Y = E\{H^{-1}(V) \mid Y, D, X, \widehat{G}\}$ for the conditional mean reference correction. The randomized estimator converges to the same completion mean as its Rao-Blackwellized form.

Proposition 2 (index projection plus correction). *With the population treatment nuisance, the population completion target is*

$$\theta_{\mathrm{SUDO}} = \frac{E\{R_D \widehat{\eta}(D, X)\}}{E(R_D^2)} + \frac{E(R_D \mu_Y)}{E(R_D^2)}. \tag{5}$$

Proof. Substitute $S = \widehat{\eta}(D, X) + \mu_Y$ into the population FWL numerator and separate its two terms. If $\widehat{\eta}(D, X) = \widehat{\alpha} D + \widehat{f}(X)$, then $E\{R_D \widehat{f}(X)\} = 0$ and $E(R_D D) = E(R_D^2)$, so the first term reduces to $\widehat{\alpha}$. □

SUDO therefore projects the fitted index onto the residualized treatment and applies an outcome-informed correction. For a structured additive fit the projection is its treatment coefficient. The correction can help when treatment regularization shrinks the index projection, hurt through covariate-direction error, or vanish. It is not a generic improvement over a direct coefficient. For binary logit, the correction reduces to an entropy-weighted residual moment.

3.2 Where orthogonality stops

The orthogonality established above protects $\widehat{\theta}$ from error in the adjustment nuisances but says nothing about the imputation model, whose error is baked into the completed data S rather than differenced away. The imputation model sits outside the orthogonality guarantee because its error is built into the outcome supplied to DML.

Here, a *black-box imputation model* means a flexible distributional learner that supplies a pointwise scalar index and conditional CDF. It need not parameterize or expose a treatment coefficient. The count validation isolates D from the joint X interaction block. An arbitrary probability vector without a scalar index on the target scale remains outside scope. Flexible imputation models are cross-fit so each supplied index is an out-of-fold prediction.

Either way $\hat{\eta}$ is fixed with respect to the completion, while $Y \mid D, X$ is generated by the truth. Since S is $\hat{\eta}$ plus truncated noise,

$$E[S \mid D, X] = \psi(\hat{\eta}, \eta), \quad \psi(v, e) = v + F(e)\,\mu_+(v) + (1 - F(e))\,\mu_-(v) \tag{6}$$

with $\mu_\pm$ the truncated-noise mean corrections. The correctly centered draw is exact, $\psi(e, e) = e$, the correct-model surrogate identity in D. Liu and Zhang (2018) and Cheng, Wang, and Zhang (2021), but index error passes through with a factor strictly below one,

$$c(e) = \left.\frac{\partial \psi}{\partial v}\right|_{v=e} = 1 - f(-e)\,(\mu_+(e) - \mu_-(e)), \qquad f = F' \tag{7}$$

Read c as the fraction of index error that survives the truncation. With no truncation the surrogate would be index plus noise and its mean would move one-for-one with $\hat{\eta}$, giving $c = 1$; the observed category drags it back, and c is what is left.

Under a logistic link, c rises from $1 - \log 2 = 0.31$ at $e = 0$ toward one in the tails (Figure 1). In fact,

$$c(e) = 1 - H(F(e)), \qquad H(p) = -p \log p - (1 - p) \log(1 - p), \tag{8}$$

so self-correction equals outcome entropy. An uncertain label sharply constrains the latent utility and corrects more of the fitted-index error. A nearly deterministic label adds little information and corrects almost none.

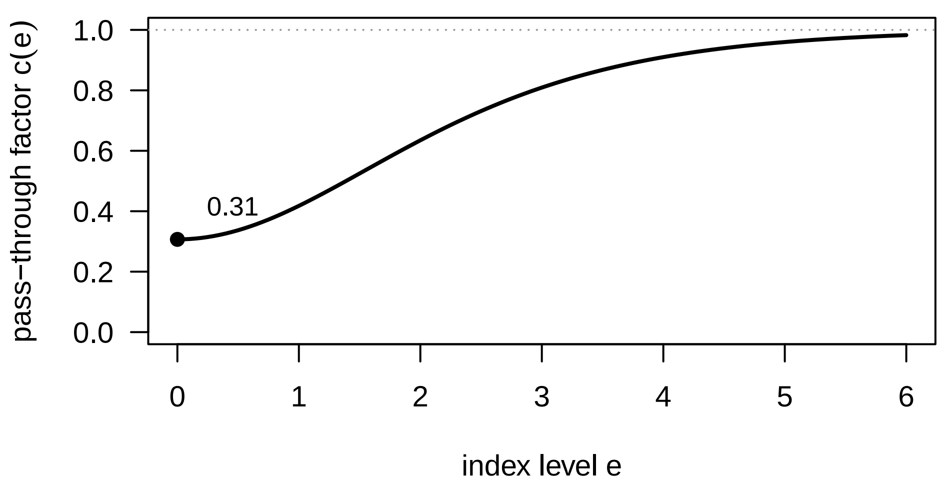

Figure 1: Fraction of fitted-index error that survives completion under a logistic link. An informative label supplies the strongest correction near the center; the correction fades as the label becomes nearly deterministic.

Fitted-index error enters the SUDO target at first order. If the effective treatment component is displaced by δ_D, the leading bias is approximately $\bar{c}_1 \delta_D$, where $\bar{c}_1$ averages $c(e)$ over the treatment-relevant index distribution. Error in the covariate component can also survive because completion is nonlinear in the fitted index. Partialling-out does not erase either channel. Target-level bias, not predictive fit alone, is therefore the relevant validation criterion.

4 Machine-learning imputation models

A parametric distribution model supplies both an index and a covariance matrix, so proper draws are immediate. A black-box learner supplies neither a treatment coefficient nor a joint parameter covariance. We therefore ask only for pointwise index and CDF predictions and handle structure and inference outside the learner.

For the validated binary experiments, we restrict the imputation index to $\hat{\alpha} D + \hat{f}(X)$. A GAM, MARS bases built from X only, and neural-network backfitting all implement this structure. The restriction prevents treatment-by-covariate interactions, but it does not guarantee a good fit or a small target bias. Two other partially linear learners failed the bias criterion, including one tuned by predictive risk.

When a learner has no usable covariance, the full-pipeline bootstrap resamples the observations and reruns the entire procedure, including the imputation learner, folds, completions, and adjustment nuisances. A within-fold refit misses important sources of variation. In the validated binary designs, normal intervals based on the full-pipeline bootstrap standard deviation attain nominal coverage.

These results support a workflow, not a blanket theorem for arbitrary learners: impose a scalar-index structure suited to the estimand, tune on target-level simulation bias, and require coverage validation before promotion.

4.1 Scale-calibrated SUDO

Regularization may preserve the shape of a fitted index while shrinking its location and scale. We correct that low-dimensional error before completion. Let a distribution learner supply an out-of-fold logit index $\hat{\eta}(D, X)$; the estimator never reads a treatment coefficient from the learner. On the training observations for a held-out fold, fit the logistic recalibration

$$P(Y = 1 \mid \hat{\eta}) = \Lambda(a + b\hat{\eta}) \tag{9}$$

and apply $\tilde{\eta} = a + b\hat{\eta}$ to the held-out index. Treatment and its residual do not enter Equation 9. This is ordinary scalar recalibration, not treatment targeting.

For binary logit, write $p = \Lambda(\tilde{\eta})$ and $h(p) = -p \log p - (1 - p) \log(1 - p)$. The Rao-Blackwell surrogate is

$$\tilde{S} = \tilde{\eta} + \frac{h(p)\{Y - p\}}{p(1 - p)}. \tag{10}$$

It equals the conditional mean of a truncated-logistic surrogate and removes completion Monte Carlo noise. Randomized completion remains available by drawing the logistic reference quantile inside the fitted Bernoulli CDF jump.

Algorithm 1: cross-fitted scale-calibrated SUDO.

1. Split the observations into K folds. Fix all tuning rules before the confirmatory sample is analyzed.
2. For each held-out fold, fit the distribution learner and treatment nuisance $m(X) = E(D \mid X)$ without that fold. Obtain scalar-index predictions for the training and held-out observations.
3. Estimate (a, b) from Equation 9 on training observations and apply the fitted map to the held-out index.
4. Construct training and held-out surrogates with Equation 10. Fit $\ell(X) = E(\tilde{S} \mid X)$ on the training observations and predict it in the held-out fold.
5. Stack the held-out predictions and compute Equation 4 with $S = \tilde{S}$. For randomized completion, repeat steps 4 and 5 for each draw before pooling the treatment estimates.
6. For a flexible distribution learner, resample the observations and rerun every preceding step. Use the standard deviation of the bootstrap estimates in the normal interval $\hat{\theta} \pm 1.96\widehat{\mathrm{sd}}_*$.

The reason for the calibration is exact in a useful special case. If, on a new population, $\eta_0(D, X) = a_0 + b_0\hat{\eta}(D, X)$ and scalar recalibration consistently estimates (a_0, b_0), then Equation 10 gives $E(\tilde{S} \mid D, X) = \eta_0(D, X)$. Population FWL therefore recovers θ_0. With a remaining error $r(D, X)$, bias depends on its projection onto $D - m_0(X)$ after completion. Scalar calibration does not control an arbitrary r and is not a substitute for target-level validation.

An optional affine fluctuation adds a treatment-residual direction to Equation 9. It is useful as a sensitivity analysis, but it partly targets the treatment effect before completion. The scale-only form is the cleaner test of whether the surrogate and FWL steps add information.

5 Simulation study

The simulations draw independent standard-normal covariates, assign treatment from those covariates, and coarsen a partially linear latent utility:

$$\begin{gathered} X_1, \ldots, X_p \overset{\text{iid}}{\sim} N(0, 1), \qquad D \mid X \sim \text{Bernoulli}(\Lambda(\pi(X))), \qquad Z = \theta_0 D + g_0(X) + \varepsilon, \\ \varepsilon \sim F, \qquad Y = j \iff \kappa_{j-1} < Z \leq \kappa_j, \end{gathered} \tag{11}$$

The correctly specified one-covariate design isolates inference. The five-covariate machine-learning design uses

$$\pi(X) = X_4 + \cos X_5, \qquad g_0(X) = X_1^2 + \sin X_2 + 0.5X_3,$$

with binary or three-category ordinal outcomes. Table 9 lists the full designs.

The coverage target is 0.95. Standard deviation (SD) is run-to-run variability; mean standard error (mean SE) is the average reported uncertainty. The tables are generated from committed result files.

Proper draws are what deliver coverage. With a simple binary design and a correct parametric imputation model, only the proper scheme reaches nominal coverage; the naive interval covers 0.80 and the improper 0.92 (Table 1).

Table 1: Inference schemes, binary outcome, $n = 5000$, $\theta_0 = 1.5$, 500 replications. B is the number of surrogate draws.

Scheme	Bias	SD	Mean SE	Coverage
Naive single draw	+0.003	0.083	0.055	0.804
Improper, $B = 25$	+0.004	0.071	0.067	0.924
Proper, $B = 25$	+0.005	0.072	0.076	0.970

Ordinal outcomes. With three categories and machine-learning adjustment, SUDO meets the bias criterion with valid coverage (Table 2). The outcome nuisance $E[S \mid X]$ must be refit for each ordinal completion; reusing one fit inflated the within-draw variance. The corrected procedure has bias 0.016 and coverage 0.960 in 200 replications.

Table 2: Ordinal outcome, $J = 3$, $\theta_0 = 1.0$. Parametric arm 500 replications; corrected spline + ML arm 200, with the outcome nuisance refit per completion.

Configuration	Bias	SD	Mean SE	Coverage
Parametric, $n = 3000$	+0.002	0.077	0.083	0.976
Spline + ML nuisances, $n = 2000$	+0.016	0.104	0.114	0.960

Direct and tailored comparisons. SUDO does not dominate a correct outcome-specific estimator (Table 3). In the correct ordinal parametric design, SUDO and the direct cumulative-link coefficient agree. Under a cloglog truth, SUDO, the tailored orthogonal score, and the direct cloglog coefficient have indistinguishable bias. In the high-dimensional binary design, SUDO attenuates treatment-regularization bias but does not remove it. These comparisons define the method's scope rather than rank efficiency.

Table 3: Direct and tailored-score comparisons. The table reports the validated common-sample cells most relevant to the scope claim.

Design	Estimator	Bias	SD
Ordinal, correct	Direct	+0.012	0.085
Ordinal, correct	SUDO	+0.011	0.084
Binary high-dimensional, pen	Tailored orthogonal	+0.029	0.116
Binary high-dimensional, pen	SUDO	-0.105	0.110
Binary cloglog, correct link	SUDO	+0.004	0.081
Binary cloglog, correct link	Tailored orthogonal	+0.004	0.081
Binary cloglog, correct link	Direct	+0.003	0.079

Link misspecification. Generating Gumbel noise but assuming a logistic link substantially biases the latent-scale effect (+0.77 for binary, +0.58 for ordinal outcomes, with zero coverage). The correct link removes this bias (bias < 0.005). Conditional second-moment drift in the surrogate residual flags the wrong fit but is not specific to the link, so we use it as a diagnostic rather than a formal test (D. Liu and Zhang 2018).

Partially-linear imputation learners. GAM, MARS, and neural backfitting fit the same $\alpha D + f(X)$ structure. All three pass the prespecified bias criterion, and full-pipeline-bootstrap normal coverage ranges from 0.94 to 0.98 (Table 4). With 50 replications, these differences are too small to rank the learners.

Table 4: Partially-linear imputation learners with full-pipeline bootstrap, $n = 2000$, 50 replications, 50 outer resamples, and 15 surrogate completions per resample.

Learner	θ_0	Bias	SD/Mean SE	Normal coverage	Percentile coverage
GAM	1.5	+0.010	0.948	0.980	0.940
GAM	3.0	+0.043	0.960	0.940	0.880
MARS	1.5	-0.009	0.978	0.960	0.920
MARS	3.0	-0.010	0.945	0.960	0.940
Neural backfit	1.5	-0.005	0.978	0.940	0.940
Neural backfit	3.0	+0.007	0.998	0.980	0.960

Partial linearity is necessary in these experiments but not sufficient. Two other partially linear learners fail the bias screen, including one selected by cross-validated predictive risk. Prediction error is therefore a diagnostic, not a tuning target for the treatment coefficient. The validated configurations were screened on target-level Monte Carlo bias and evaluated on separate seeds; a data-only selection rule remains open.

Coefficientless additive learner. We next fit the binary index with GAMSEL (Chouldechova and Hastie 2015), constraining D to be linear while allowing each X_j to enter linearly, smoothly, or not at all. Estimation uses only index predictions. In the three $p = 80$ additive designs, the full affine-calibrated SUDO pipeline passes every prespecified bias, coverage, SD-to-SE, and bootstrap-success gate (Table 5). Ordinary SUDO retains material regularization bias.

Table 5: Coefficientless GAMSEL learner with treatment coefficient 0.75. Full-affine results use 50 fresh datasets and 99 full-pipeline bootstrap resamples.

Design	Ordinary bias	Calibrated bias	Coverage	SD/Mean SE
Randomized	-0.057	-0.012	0.980	0.917
Confounded	-0.076	-0.014	0.940	0.976
High signal	-0.102	-0.009	0.960	1.020

Table 6: Scale-only mechanism screen on separate seeds. Entries are bias over 10 fresh datasets.

Design	Direct read-off	Scale-calibrated SUDO
Randomized	+0.019	+0.014
Confounded	+0.039	+0.009
High signal	+0.009	-0.009

The scale-only ablation isolates SUDO's contribution. Its biases are $+0.014$, $+0.009$, and -0.009 across the three designs. Under confounding, reading a coefficient from the scale-calibrated index has bias $+0.039$, while completion and FWL reduce it to $+0.009$. Tilt-only calibration fails under confounding and high signal. Thus treatment-directed targeting is not needed for the point result. Scale-only coverage has not yet been validated, so Table 5 and Table 6 report full-affine coverage and scale-only bias separately. At $p = 500$, full-affine calibrated SUDO has bias -0.048 and fails the point gate. This is the observed boundary, not a promoted design.

With continuous treatment, bias remains below 1.5% of the truth in the binary and ordinal designs. Strong-signal coverage is less stable, reinforcing the need to rerun the full-pipeline bootstrap rather than transfer an uncertainty result from a binary-treatment experiment.

5.1 Count oracle bridge

We isolate count completion from fitted-learner error by supplying the true scalar index and count CDF. The six prespecified cells cross Poisson and negative-binomial laws, binary and continuous treatments, and two signal strengths where applicable. At $n = 2000$, with 100 replications, $B = 25$ randomized completions, five folds, and 99 bootstrap resamples, both analytic Rao-Blackwell and randomized completion pass every bias, coverage, and SD-to-SE gate (Table 7). Their paired differences are within two completion Monte Carlo standard errors in every cell.

Table 7: Count oracle bridge with the true index and CDF supplied. Entries pair Rao-Blackwell and randomized completion. Each cell uses $n = 2000$, 100 replications, five folds, and 99 bootstrap resamples; randomized SUDO uses $B = 25$.

Design	θ_0	Bias (RB / randomized)	SD/SE (RB / randomized)	Coverage (RB / randomized)
Continuous, Poisson	$\log(1.5)$	-0.001 / -0.001	0.989 / 0.989	0.970 / 0.970
Continuous, NB	$\log(1.5)$	+0.001 / +0.001	1.046 / 1.042	0.940 / 0.940
Continuous, Poisson	$\log(2)$	+0.001 / +0.001	0.921 / 0.925	0.980 / 0.980
Continuous, NB	$\log(2)$	-0.001 / -0.001	0.963 / 0.962	0.940 / 0.950
Binary, Poisson	$\log(2)$	-0.001 / -0.001	1.127 / 1.120	0.910 / 0.920
Binary, NB	$\log(2)$	-0.002 / -0.002	1.114 / 1.105	0.920 / 0.920

This bridge validates the count completion and bootstrap when the supplied distribution is correct. It does not validate a fitted count learner. The constrained-XGBoost point screen failed its promotion gate, so fitted count support and the prespecified bikeshare application remain outside the paper's claims.

6 Application: volatile acidity and wine quality

We apply ordinal SUDO to the UCI wine-quality data (Cortez et al. 2009). Quality is the median ordinal score from blind tasters. We analyze the 1,599 red and 4,898 white wines separately; observed scores range from 3 to 8 for red wine and 3 to 9 for white. The treatment is standardized volatile acidity, an acetic-acid fault associated with vinegar-like aroma. The adjustment set is the other ten standardized physicochemical measurements.

The data are observational. A causal reading requires the measured chemistry to be a sufficient pre-treatment adjustment set and winemaking practice to affect quality only through that chemistry and volatile acidity (Figure 2). We do not claim that these assumptions are established. The analysis illustrates the method on a genuinely ordinal outcome.

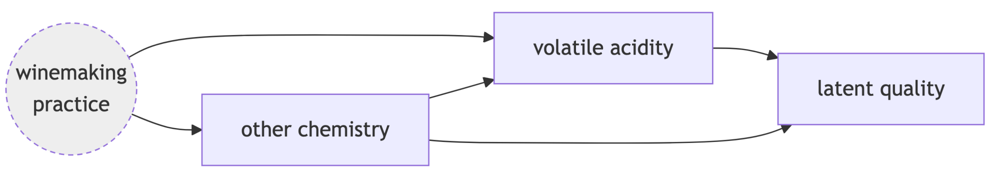

Figure 2: Assumed application DAG. Volatile acidity (the treatment) affects latent quality directly; unobserved winemaking practice confounds both through the measured chemistry, the adjustment set.

Volatile acidity retains 75% of its marginal standard deviation after adjustment in red wine and 94% in white wine. Both imputation models are proportional-odds logits with volatile acidity entering linearly. One uses the adjustment variables linearly; the other uses a three-degree-of-freedom natural spline for each variable, with no treatment-by-covariate interactions. Ordinal SUDO uses 50 proper draws, five-fold cross-fitted GAM nuisances, and Rubin pooling. The two specifications give similar estimates (Table 8).

Because volatile acidity is standardized, the coefficients are per one standard deviation, about 0.18 g/dm^3 for red wine and 0.10 for white. A one-SD increase multiplies the odds of scoring above any fixed quality level by about 0.59 for red wine and 0.64 for white wine. The direct cumulative-link coefficient is included as a model-based comparator.

Table 8: Effect of a one-SD increase in volatile acidity on latent wine quality. Model-based coefficient from the linear cumulative-link fit (standard error in parentheses); SUDO estimates with 95% intervals in brackets.

Wine	Overlap $R^2(D \sim X)$	Cumulative-link	SUDO (linear)	SUDO (spline)
Red	0.44	-0.61 (0.07)	-0.52 [-0.68, -0.36]	-0.52 [-0.69, -0.35]
White	0.12	-0.50 (0.03)	-0.45 [-0.58, -0.33]	-0.45 [-0.58, -0.32]

The negative sign is stable when any one adjustment covariate is removed. An omitted-variable calibration suggests that a confounder would need to explain roughly 15% of the residual treatment and outcome variation for red wine and 10%

for white wine to move the estimate to zero (Cinelli and Hazlett 2020). Because that calibration was derived for linear regression with an observed outcome, we report it as a heuristic rather than a formal SUDO bound.

The completion itself provides substantial self-correction in this application: the fitted pass-through averages 0.21 for red wine and 0.17 for white wine. This does not establish robustness to an arbitrary wrong imputation model, but it shows why the ordinal categories carry useful information beyond the fitted index.

7 Discussion

SUDO turns a discrete-outcome problem into a familiar continuous partialling-out problem. The fitted distribution identifies the target scale; randomized completion restores that scale in conditional mean; DML then removes flexible covariate effects. This separation is the main value of the method.

The method does not dominate a correct direct or tailored estimator. Its advantage is a common interface, especially for ordinal outcomes for which a tailored orthogonal score is not readily available. Proper draws or a full-pipeline bootstrap are part of the estimator, not optional refinements.

The same separation also exposes the central risk. DML orthogonality protects the adjustment regressions, but not the distribution learner that creates the surrogate. Partial linearity prevents treatment-by-covariate interactions but does not guarantee small treatment bias. Each new learner and outcome family therefore needs target-level simulation and coverage checks.

Scalar recalibration addresses one common failure mode: a learner may recover the index shape but shrink its location and scale. In the additive GAMSEL experiment, scale-only recalibration followed by SUDO passes all point-bias gates without using a treatment direction. The full affine version also passes full-pipeline coverage. Neither result implies protection from arbitrary index error; the $p = 500$ boundary fails.

The validated contribution is binary and ordinal, including a calibrated coefficientless binary learner. The count oracle bridge shows that completion extends cleanly to Poisson and negative-binomial laws, but the fitted count learner did not pass promotion. Full coverage validation of scale-only calibrated SUDO remains the next binary inference check.

8 Data and software

The wine-quality data come from the University of California, Irvine Machine Learning Repository (Cortez et al. 2009). The `sudo` Python package provides the parametric estimator for binary and ordinal outcomes, including proper draws and the per-completion ordinal outcome-nuisance refit. The partially-linear machine-learning imputation learners and the full-pipeline bootstrap are implemented in R, which is also where every method is prototyped and validated before release. Count and flexible ordinal adapters remain R-only until their fitted-learner confirmation stages pass. Scale-calibrated SUDO also remains R-only pending its own coverage stage. The open-source repository contains the code and committed result files behind every table and figure under the MIT License: https://github.com/bgreenwell/sudo.

9 Declaration of artificial intelligence use

We used generative artificial intelligence (AI) to assist with code and summaries of reference materials. The authors reviewed and validated all outputs, retain full ownership of the final manuscript, and accept sole responsibility for any errors.

10 Appendix: simulation and application details

Each design draws a latent utility $Z = \theta_0 D + g_0(X) + \varepsilon$ and converts it to an observed category. Most designs use logistic ε; the exceptions are specified below. The designs vary the covariate structure, cutpoints, and fitted models; covariates remain standard normal throughout. The `R/` and `R/results/` directories contain the full data-generating code and committed per-cell results.

10.1 Data-generating processes

All designs instantiate Equation 11 with one of two covariate structures:

- **S1**, one covariate: $p = 1$, $\pi(X) = 0.8X$, $g_0(X) = X$. Every model in play is correctly specified, so only the pooling is under test.
- **S5**, five covariates: $p = 5$, $\pi(X) = X_4 + \cos X_5$, $g_0(X) = X_1^2 + \sin X_2 + 0.5X_3$. The outcome structure is nonlinear in X, and treatment depends on covariates absent from g_0.

Table 9: Data-generating processes. κ lists the interior cutpoints, so (0) is binary and a pair gives $J = 3$.

Design	Structure	F	κ	θ_0	n	Replications
Inference ladder (Table 1)	S1	logistic	(0)	1.5	5000	500
Ordinal parametric (Table 2, top)	S1	logistic	$(-1, 1)$	1.0	3000	500
Binary ML	S5	logistic	(0)	0.5, 3	1000, 2000	200
Ordinal flexible adjustment (Table 2, bottom)	S5	logistic	$(0, 2)$	1.0	1000, 2000	200
Link misspecification	S1	Gumbel	(0) or $(0, 2)$	1.0	5000	200
Black-box imputation	S5	logistic	(0)	1.5, 3	2000	100
PL learner comparison (Table 4)	S5	logistic	(0)	1.5, 3	2000	50
Calibration (Table 5)	$p = 80$ additive	logistic	(0)	0.75	3000	50
DGP variation, continuous D	S5, $D = 0.7\pi(X) + N(0, 1)$	logistic	(0) or $(0, 2)$	1, 2.5	2000	200

Models fitted. The ladder uses $\mathrm{glm}(Y \sim D + X)$ as the imputation model with $B \in \{10, 25, 50\}$ draws (the table shows $B = 25$). The ordinal parametric design uses `ordinal::clm`, whose covariance covers the thresholds, with $B = 25$. The ordinal flexible-adjustment design uses parametric `clm` on D and natural-spline bases $\mathrm{ns}(X_k, 4)$ with `mgcv::gam` partialling nuisances and $B = 25$.

Link misspecification. This design retains the one-covariate structure but replaces logistic ε with an extreme-value error: Gumbel-max for the binary case, matching `glm` with a cloglog link, and Gumbel-min for the ordinal case, matching `ordinal::clm` with a cloglog link. The analyst fits either logit (misspecified) or cloglog (correct). Holding the simple covariate structure fixed isolates error in the link.

Partially-linear imputation models. The original black-box result uses $\hat{\eta} = \hat{\alpha}D + \hat{f}(X)$ with $\hat{f}$ a single-hidden-layer neural net (`nnet`, `size = 4`, `decay = 0.3`, five backfitting iterations, selected on target-level Monte Carlo bias) and 100 full-pipeline bootstrap resamples. The common comparison uses 50 resamples per arm. The GAM includes a linear treatment term and smooths of X. MARS builds degree-two bases from X only before a joint logistic fit with mandatory linear D.

Calibration experiment. The calibration designs use $p = 80$ independent standard-normal covariates and

$$g_0(X) = 0.65X_1 + 0.55(X_2^2 - 1) + 0.45X_3 - 0.35(X_4^2 - 1) + 0.30X_5 - 0.25X_6.$$

Treatment is $N(0, 1)$ in the randomized design. In the confounded design, $D = m_0(X) + N(0, 1)$ with $m_0(X) = 0.50X_1 - 0.40X_3 + 0.30(X_2^2 - 1)$. The high-signal design uses $D \sim N(0, 4)$. Outcomes follow $Y \sim \mathrm{Bernoulli}[\Lambda\{0.75D + g_0(X)\}]$. GAMSEL uses cubic candidate smooths for X, a linear treatment term, and cross-validated binomial deviance. Five outer and five inner folds are used. The point ablation uses separate seeds from tuning and coverage.

Wine application. Red and white wines are fit separately. The outcome levels are 3–8 for red wine and 3–9 for white. Volatile acidity is the standardized treatment. The ten standardized adjustment variables are fixed acidity, citric acid, residual sugar, chlorides, free and total sulfur dioxide, density, pH, sulphates, and alcohol. The proportional-odds logit full model is `ordinal::clm` with index $\alpha D + \beta^\top X$ (linear) or $\alpha D + \sum_k \mathrm{ns}(X_k, 3)$ (spline). Partialling nuisances $E[S \mid X]$ and $E[D \mid X]$ use `mgcv::gam` (Gaussian for continuous D), with five-fold cross-fitting, $B = 50$ proper draws, and Rubin pooling.

10.2 Nuisances, cross-fitting, and draws

Partialling nuisances $\widehat{\ell}(X) = E[S \mid X]$ and $\widehat{m}(X) = E[D \mid X]$ are `mgcv::gam` throughout, fit with five-fold cross-fitting. Proper draws sample the full-model parameters from $N(\widehat{\beta}, \widehat{\mathrm{Cov}}(\widehat{\beta}))$ (thresholds included for `clm`) before each completion. Diagnostic draws use a random number generator (RNG) stream independent of the estimation draws.

Binary experiments found per-completion refitting of $\widehat{\ell}$ negligible, so the binary estimator reuses a plug-in fit. The ordinal estimator refits $\widehat{\ell}$ for every completion because reuse inflated its within-draw variance.

10.3 Monte Carlo precision

Coverage is a binomial proportion, so its Monte Carlo standard error at a true 0.95 is about $\sqrt{0.95 \cdot 0.05/R}$: 0.010 at $R = 500$ replications, 0.015 at $R = 200$, and 0.022 at $R = 100$. Per-cell bias and coverage Monte Carlo standard errors are in the committed result files.

11 References

Barnard, John, and Donald B. Rubin. 1999. "Small-Sample Degrees of Freedom with Multiple Imputation." *Biometrika* 86 (4): 948–55. https://doi.org/10.1093/biomet/86.4.948.

Cheng, Chao, Rui Wang, and Heping Zhang. 2021. "Surrogate Residuals for Discrete Choice Models." *Journal of Computational and Graphical Statistics* 30 (1): 67–77. https://doi.org/10.1080/10618600.2020.1775618.

Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. "Double/Debiased Machine Learning for Treatment and Structural Parameters." *The Econometrics Journal* 21 (1): C1–68. https://doi.org/10.1111/ectj.12097.

Chouldechova, Alexandra, and Trevor Hastie. 2015. "Generalized Additive Model Selection." *arXiv Preprint arXiv:1506.03850*. https://doi.org/10.48550/arXiv.1506.03850.

Cinelli, Carlos, and Chad Hazlett. 2020. "Making Sense of Sensitivity: Extending Omitted Variable Bias." *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 82 (1): 39–67. https://doi.org/10.1111/rssb.12348.

Cortez, Paulo, António Cerdeira, Fernando Almeida, Telmo Matos, and José Reis. 2009. "Modeling Wine Preferences by Data Mining from Physicochemical Properties." *Decision Support Systems* 47 (4): 547–53. https://doi.org/10.1016/j.dss.2009.05.016.

Drukker, David M., and Di Liu. 2022. "Finite-Sample Results for Lasso and Stepwise Neyman-Orthogonal Poisson Estimators." *Econometric Reviews* 41 (9): 1047–76. https://doi.org/10.1080/07474938.2022.2091363.

Dunn, Peter K., and Gordon K. Smyth. 1996. "Randomized Quantile Residuals." *Journal of Computational and Graphical Statistics* 5 (3): 236–44. https://doi.org/10.1080/10618600.1996.10474708.

Greenwell, Brandon M., Andrew J. McCarthy, Bradley C. Boehmke, and Dungang Liu. 2018. "Residuals and Diagnostics for Binary and Ordinal Regression Models: An Introduction to the sure Package." *The R Journal* 10 (1): 381–94. https://doi.org/10.32614/RJ-2018-004.

Liu, Dungang, and Heping Zhang. 2018. "Residuals and Diagnostics for Ordinal Regression Models: A Surrogate Approach." *Journal of the American Statistical Association* 113 (522): 845–54. https://doi.org/10.1080/01621459.2017.1292915.

Liu, Molei, Yi Zhang, and Doudou Zhou. 2021. "Double/Debiased Machine Learning for Logistic Partially Linear Model." *The Econometrics Journal* 24 (3): 559–88. https://doi.org/10.1093/ectj/utab019.

Rubin, Donald B. 1987. *Multiple Imputation for Nonresponse in Surveys*. New York: John Wiley & Sons. https://doi.org/10.1002/9780470316696.